

WQU Capstone

Assignment: Group Formation & Topic Selection

This form **must be submitted in your online course room immediately after the course begins**. Failure to submit your group formation and topic selection will result in cancellation from the course at the start of Week 2. If you are proposing your own topic, you are required to send this form to Student Support using the Help Center chatbot prior to the M1 deadline for instructor approval.

Full Legal Name	Davide Freni (devide.freni@gmail.com)
Email Address	

Name of group member(s) (*If working alone, input "Solo")	Vicky Singh (ivickysingh@yahoo.com) Kenneth Kariuki (kenkarish1935@gmail.com)
--	--

Select your track/topic from the dropdown menu	8. Machine Learning (Deep) Investment Strategies
If response above is "Other", insert your proposed topic here	

This section **is required** to provide additional information regarding your topic as this will help instructors determine if they would be a good fit to mentor your group for the duration of the project. If you opted to propose your own topic, this information will be used for approval purposes (ie. whether you have access to necessary data, etc.).

This study will introduce an adaptable portfolio allocation system that combines machine learning, deep learning, and artificial intelligence methods to modify asset allocations. The system will collect and standardize different asset categories, economic metrics, and supplementary data while recognizing influential characteristics to reduce future biases. Agents using reinforcement learning will refine strategies while adhering to operational limitations, and sequence-sensitive models, including transformers and LSTMs, will predict fluctuations in returns and risks. To ensure institutional standards of transparency, resilience, and regulatory compliance, rigorous walk-forward testing, strict risk controls, and strong model governance—including explainability and performance monitoring—will be crucial. This will allow for real-time, production-ready portfolio management.

Statement of integrity: *By typing my name in the text box below, I confirm that I am willingly entering into group formation with the students listed above and have confirmed the common interest in the topic for this project. I understand that failure to collaborate effectively with the other students of the group will result in grades of 0 and automatic isolation. If I am proposing my own topic I am required to submit this form to the Help Center chatbot prior to the course start date for instructor approval.*

Full Legal Name	Davide Freni
Date Submitted	September 14 th , 2025